Simulate a circuit and practice soldering

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Test a circuit representation before making permanent connections.

SUCCESS CRITERIA - I can...

1. Build a complete battery-switch-motor circuit in the simulation.
2. Explain an open versus closed path.
3. Document a supervised solder-practice check.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “open circuit” and “short circuit” to explain your answer. Sentence starters are optional.

open circuit: a break that prevents current through that path

short circuit: an unintended low-resistance path bypassing a component

insulation: material that limits unwanted electric conduction

01 Warm-up: bring back an earlier idea

A car prototype is ready for a planned **5 m** powered run. **Name one safety check** to make before turning on the power and one measurement needed to judge the run.

Word help: evaluate = check a claim against evidence and conditions

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $I = \Delta q/\Delta t$; $I = \mathbf{0}$ in an ideal open circuit.
Use the assigned simulation and teacher solder procedure. Disconnect power; hot irons/joints stay in the designated work area.

Worked reasoning: In a simple car circuit, battery positive connects through the switch and motor back to battery negative. A wire directly across the battery bypasses the motor and can create a hazardous short circuit.

Your turn: Draw the intended closed loop and circle a place where an open switch safely breaks it.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

1. **Draw your simulated closed circuit** and an open-switch version.
2. **Predict which runs the motor** and why.

Sentence starter: I predict … . I used … to make this prediction because … .



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: 1 2 3

04 Apply the physics

In a circuit simulation, a **3 V** supply delivers **0.20 A** to a motor. **Label voltage** and current and **calculate input power**. Make one approved change in the simulation and **record what you actually observe**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Make a two-column record: practice-joint observation and next improvement. Include teacher feedback on the joint and tool handling.

Sentence starter: I notice … . This supports … because … .

Observed joint / handling	Teacher feedback	Next time
___	___	___
___	___	___

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Parallel branches draw **0.4 A** and **0.6 A**. **Find the current supplied** to their junction.
Draw the stated circuit and **label quantities relevant** to the question.

Word help: current = rate of charge flow

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Before soldering a circuit connection, should power be connected or disconnected? **State one reason**.

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Draw two circuit faults on the blank circuit diagrams: (A) one disconnected lead; (B) a wire that bypasses the motor and connects across the battery. **Explain the expected behavior** and **name a teacher-approved first check** with power disconnected. These are drawings, not faults to build.

Sentence starter: I predict … . I used … to make this prediction because … .

A: open lead

B: bypass

Draw each circuit fault; label the motor and battery.

Marzano check - circle: 1 2 3